

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Jeshyapriya.R(192524453) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Jeshyapriya.R(192524453)

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

Problem: We have two jugs: a 4-gallon jug and a 3-gallon jug. We need to measure exactly 2 gallons using these jugs.

Solution: Step-by-Step

Step	Action	4-Gallon Jug	3-Gallon Jug
1	Fill the 3-gallon jug	0	3
2	Pour water into the 4-gallon jug	3	0
3	Fill the 3-gallon jug again	3	3
4	Pour water into the 4-gallon jug until it is full	4	2
5	Empty the 4-gallon jug	0	2
6	Pour the remaining water into the 4-gallon jug	2	0

Result: Exactly 2 gallons of water are measured in the 4-gallon jug.

Assumptions

1. Both jugs have fixed capacities.
2. Water is available in unlimited quantities.
3. Water can be poured between jugs or emptied onto the ground.
4. No measuring devices are available.

Conclusion

The water jug problem can be solved using a **state-space search approach**, where each state represents the amount of water in the two jugs.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?

- v. What sort of agent architecture do you think is most suitable for this agent? Why?

A Mars Rover is an autonomous robot designed to explore the surface of Mars, collect samples, and send information to Earth.

i. What are the percepts for this agent?

Percepts are the information received by an agent through its sensors.

The Mars Rover receives the following percepts:

- Images of the Martian surface.
- Temperature and atmospheric conditions.
- Distance to rocks and obstacles.
- Soil and rock composition.
- Rover position and movement.
- Battery level and system status.

ii. Characterize the operating environment.

The Mars Rover operates in the following environment:

- **Partially observable:** It cannot observe the entire Martian surface at once.
- **Single-agent:** It performs its exploration independently.
- **Stochastic:** Some outcomes, such as wheel movement on loose soil, are uncertain.
- **Sequential:** Each action affects future actions.
- **Dynamic:** Environmental conditions may change.
- **Continuous:** Position, distance, temperature, and movement can vary continuously.

iii. What are the actions the agent can take?

The Mars Rover can perform these actions:

1. Move forward and backward.
2. Turn left and right.
3. Capture images.
4. Collect soil and rock samples.
5. Analyze samples using scientific instruments.
6. Avoid obstacles.
7. Communicate data to Earth.
8. Recharge or conserve battery power when possible.

iv. How can one evaluate the performance of the agent?

The performance of the Mars Rover can be evaluated using:

- Number of samples collected successfully.
- Accuracy of scientific data.
- Distance explored.
- Successful obstacle avoidance.
- Efficient battery usage.
- Amount of useful data transmitted to Earth.
- Safety and reliability of operations.

v. What type of agent architecture is most suitable? Why?

Answer: Model-Based Utility Agent

A model-based utility agent is suitable because the Mars Rover operates in a partially observable and uncertain environment.

Reasons:

1. It maintains an internal model of the environment.
2. It uses sensor data to update its understanding.
3. It compares possible actions based on their benefits.
4. It selects actions that improve scientific results while considering safety, time, and energy.

Conclusion: A model-based utility agent helps the Mars Rover make intelligent decisions and explore Mars efficiently.

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

Problem:

Place eight queens on an 8×8 chessboard so that no two queens attack each other.

Rules

- No two queens can be in the same row.
- No two queens can be in the same column.
- No two queens can be on the same diagonal.

Search Strategy

The **Backtracking Search Algorithm** is suitable for solving the Eight Queens Problem.

Algorithm

1. Start with an empty chessboard.
2. Place a queen in the first row.
3. Check whether the position is safe.
4. If safe, move to the next row.
5. If no safe position exists, return to the previous row and move that queen.
6. Continue until all eight queens are placed safely.

Pseudocode

N = 8

FUNCTION Solve(row):

 IF row == N:
 PRINT board
 RETURN TRUE

 FOR col = 0 TO N-1:
 IF position (row, col) is safe:
 Place queen at (row, col)

 IF Solve(row + 1):
 RETURN TRUE

 Remove queen from (row, col)

 RETURN FALSE

```

BEGIN:
    Create an empty 8 × 8 board
    Solve(0)
END
One Valid Solution\

```

The following table gives the column position of the queen in each row.

Row	Column
1	1
2	5
3	8
4	6
5	3
6	7
7	2
8	4

The corresponding column sequence is:

[1, 5, 8, 6, 3, 7, 2, 4]

Conclusion

Backtracking systematically explores possible queen positions and removes choices that lead to conflicts. This reduces unnecessary searching and finds a valid solution.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

Problem: A customer wants to travel from one location to another using OLA Cab. The customer can choose a mini, micro, sedan, or prime cab based on comfort and price.

Type of Problem-Solving Agent

A **Goal-Based Agent with Cost Optimization** is suitable.

The agent identifies the customer's destination and selects a cab that satisfies the customer's requirements while minimizing the cost or travel time.

Steps Involved

1. **Initial state:** Customer's current location.
2. **Goal state:** Customer reaches the destination.
3. **Actions:** Select cab, book cab, and travel to the destination.
4. **State space:** Available locations, routes, and cab options.
5. **Path cost:** Cab fare, travel time, and other relevant costs.
6. **Goal test:** Check whether the customer has reached the destination.
- 7.

Pseudocode

```

BEGIN

```

```

    INPUT source
    INPUT destination
    INPUT preferred_cab
    INPUT budget

```

```

    Get available cabs

```

Get estimated fares and travel times

FOR each cab:

IF cab matches customer preference

AND fare $\leq$ budget:

Add cab to suitable_cabs

IF suitable_cabs is empty:

PRINT "No suitable cab available"

ELSE:

Select the best cab based on fare and travel time

Display cab details and estimated fare

IF customer confirms booking:

Book the selected cab

Navigate to destination

PRINT "Booking successful"

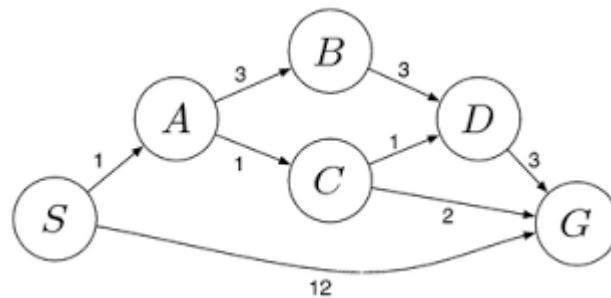
END

Conclusion

A goal-based agent with cost optimization helps customers choose a suitable cab based on their preferences, budget, and estimated travel time.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse **G** using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Problem: A logistics company needs to transport packages from the starting warehouse **S** to the destination warehouse **G** at minimum cost.

Uniform Cost Search expands the path with the lowest cumulative cost first.

Given Weighted Graph

The edges and costs shown in the figure are:

Route	Cost
S $\rightarrow$ A	1
S $\rightarrow$ G	12
A $\rightarrow$ B	3
A $\rightarrow$ C	1
B $\rightarrow$ D	3
C $\rightarrow$ D	1

$C \rightarrow G$	2
$D \rightarrow G$	3

Step-by-Step UCS Solution

Step 1: Start at S with cost 0.

Available paths:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$

Choose **S** $\rightarrow$ **A**, because its cost is 1.

Step 2: Expand A.

New paths:

- $S \rightarrow A \rightarrow B = 1 + 3 = 4$
- $S \rightarrow A \rightarrow C = 1 + 1 = 2$
- $S \rightarrow G = 12$

Choose **S** $\rightarrow$ **A** $\rightarrow$ **C**, with cost 2.

Step 3: Expand C.

New paths:

- $S \rightarrow A \rightarrow C \rightarrow D = 1 + 1 + 1 = 3$
- $S \rightarrow A \rightarrow C \rightarrow G = 1 + 1 + 2 = 4$

Choose **S** $\rightarrow$ **A** $\rightarrow$ **C** $\rightarrow$ **D**, with cost 3.

Step 4: Expand D.

New path:

- $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G = 1 + 1 + 1 + 3 = 6$

The lowest-cost remaining path to the goal is now $S \rightarrow A \rightarrow C \rightarrow G$, with cost 4.

Step 5: Select G with cost 4. Since the destination is reached with the lowest cumulative cost, the search terminates.

Final Answer

Optimal Path:

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost:

$$\text{Total Cost} = 1 + 1 + 2$$

$$\boxed{\text{Minimum Cost} = 4}$$

Conclusion

Uniform Cost Search finds the least-cost route by expanding paths in increasing order of cumulative cost. The optimal route from warehouse S to warehouse G is **S** $\rightarrow$ **A** $\rightarrow$ **C** $\rightarrow$ **G**, with a total cost of **4 units**.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem